HEALTHCARE DIGITAL TWIN · CASE COMPETITION

What if we could rehearse the surge?



HospitalFlow AI

An interactive Emergency Department twin for exploring patient flow, bottlenecks, and staffing decisions.

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THE PROBLEM

Bottlenecks emerge before dashboards explain them.

ED demand, triage capacity, clinicians and downstream beds interact as one system.

01. Demand is stochastic

Arrival surges compound waiting times as limited resources compete for attention.

02. Queues are coupled

A bottleneck at one step can influence triage, treatment and critical-care queues.

03. Policies need testing

A static census cannot show how a different staffing allocation might change future states.

OUR QUESTION: Can a virtual ED rehearse disruptions and evaluate resource choices transparently?

THE SOLUTION

From live state to counterfactual decisions.

A digital twin connects incoming events, a stateful model, metrics and intervention tests.

01

SYNTHETIC EVENTS

Seeded Poisson arrivals and patient acuity.

→

02

VIRTUAL ED STATE

Priority queues, service capacities and patient pathways.

→

03

SCENARIO ENGINE

Replays, constrained allocation search and forecasting.

→

04

HUMAN-IN-THE-LOOP UI

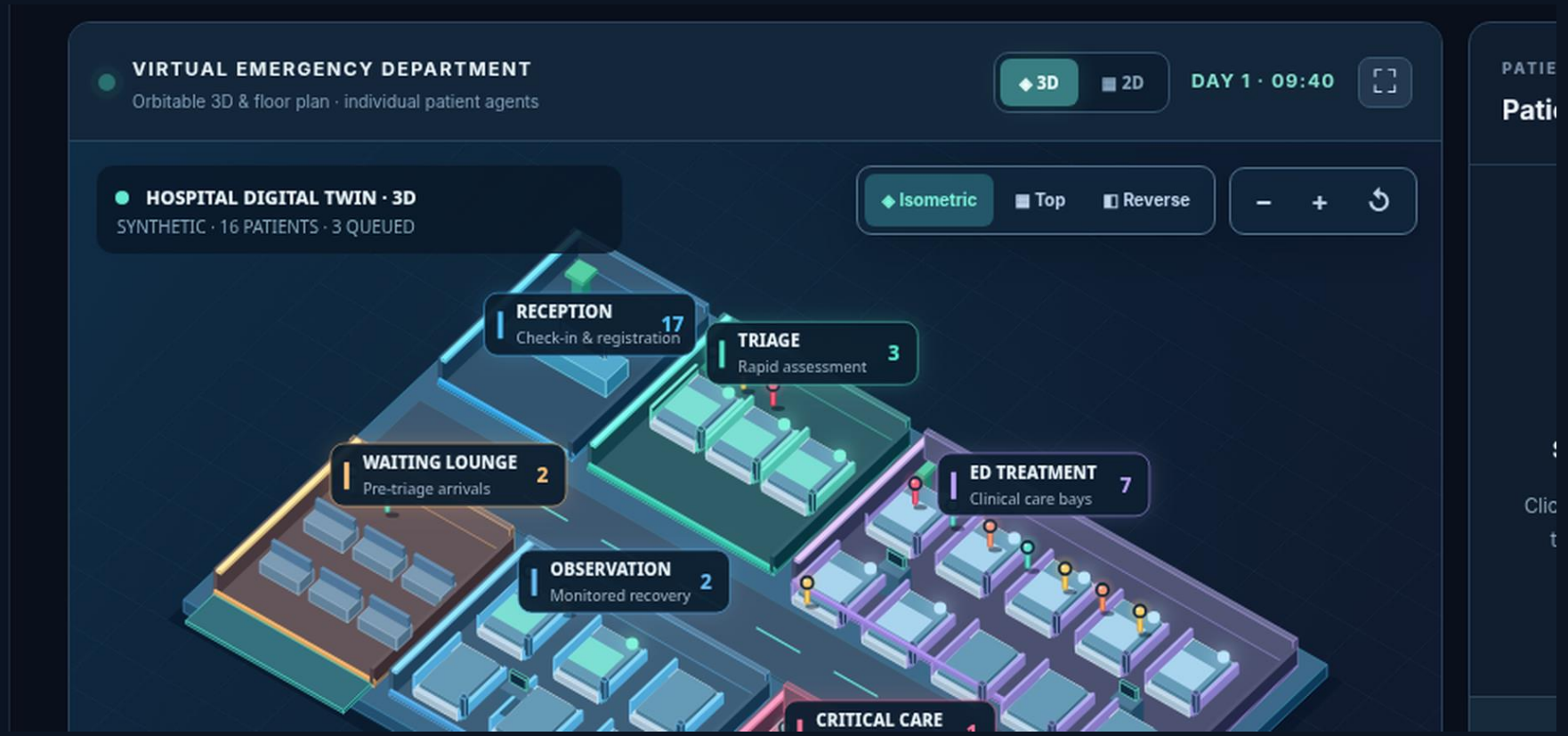
3D / 2D rooms, patient inspector and scenario controls.

FEEDBACK LOOP: USER CHANGES INPUTS → TWIN UPDATES → NEW SCENARIOS

LIVE PRODUCT

Explore a hospital in 3D — and follow every patient.

Orbit, zoom, click a patient or switch to 2D. Both views reflect the same simulation state.



3D hospital

Orbitable rooms + live beds

2D / 3D toggle

Same patients + journey trace

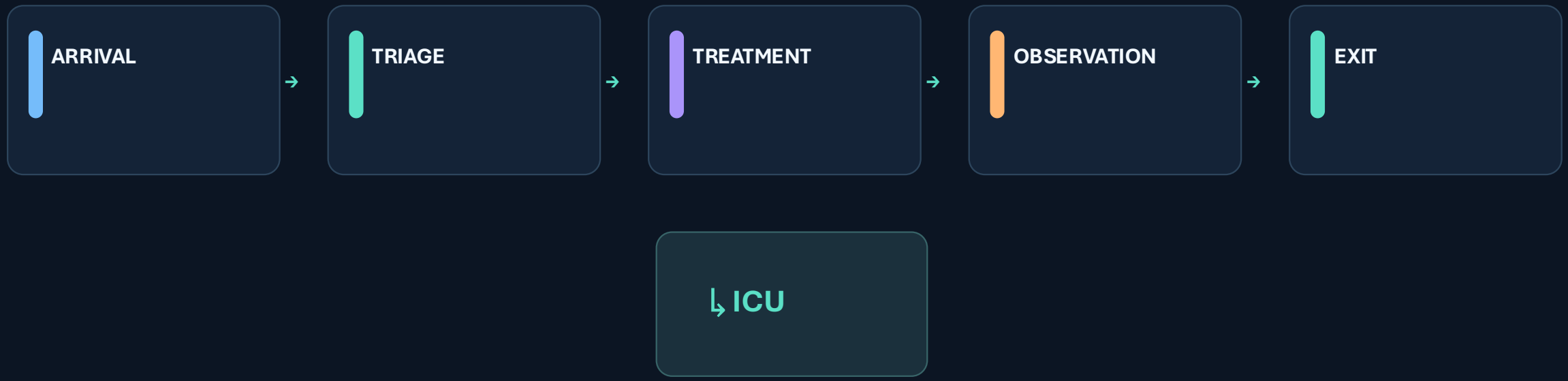
What-if controls

Staff, beds, arrivals, surge

MODEL MECHANICS

A patient-level model with constrained resources.

Each synthetic patient carries an acuity, arrival timestamp, stage and remaining service time.



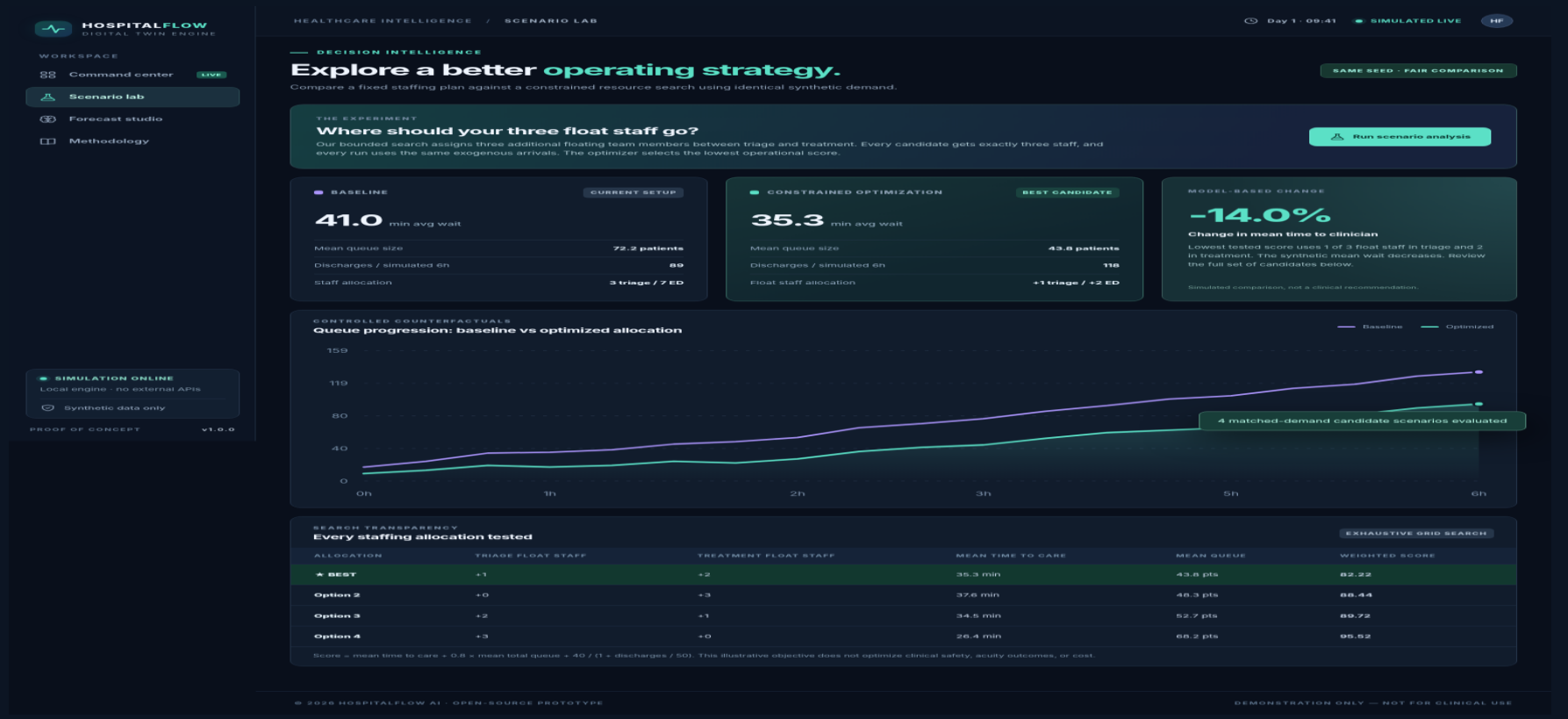
Priority: P1 first

Constraints: nurses, physicians, treatment beds, ICU beds

INTERACTIVE EXPERIMENTS

Run the same surge under different staff allocations.

All four candidate configurations are constrained to three additional float staff.



EXPERIMENT

3

floating staff

4

staff allocations

SIMULATION RESULTS

Adding flexible capacity changes modeled congestion.

Illustrative deterministic six-hour evaluation · 90-minute warmup · fixed seed 34071.

STANDARD ARRIVAL RATE

17 base arrivals / hour

BASELINE	WITH 3 FLOAT STAFF
14.36 min	8.22 min

MEAN QUEUE: 2.00 → 0.05 patients

SURGE ARRIVAL RATE

×1.8 demand multiplier

BASELINE	WITH 3 FLOAT STAFF
32.26 min	30.07 min

MEAN QUEUE: 70.11 → 42.70 patients

These are SIMULATED comparisons, not real hospital performance gains or validated clinical policies.

EXPLAINABLE OPTIMIZATION

No black-box recommendations.

Exhaustively test all valid splits of the same three floating staff.

PLAN 01

0 nurses + 3 physicians

PLAN 02

LOWEST TESTED SCORE

1 nurse + 2 physicians

PLAN 03

2 nurses + 1 physician

PLAN 04

3 nurses + 0 physicians

COMPOSITE OBJECTIVE

mean wait + 0.8 × mean queue + 40 / (1 + discharges / 50)

FORECASTING COMPONENT

A real trained model. A transparent synthetic benchmark.

Ridge regression on Fourier seasonality features; no hospital dataset or protected patient information.

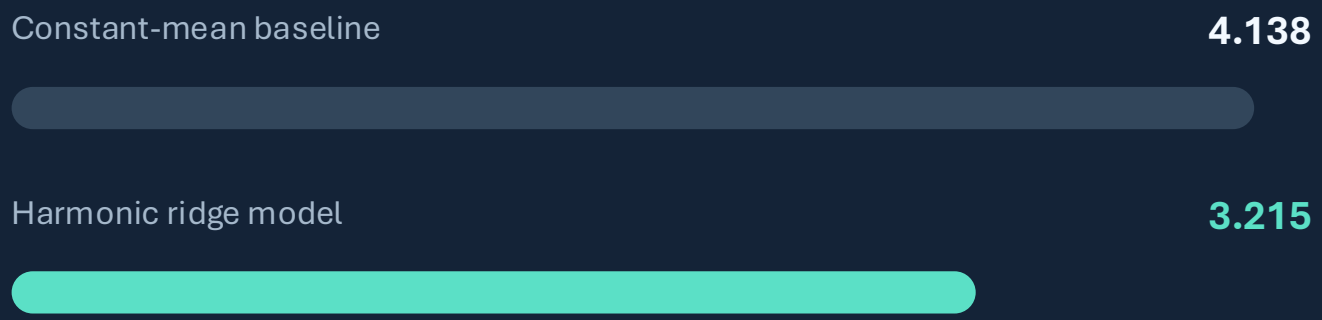
TRAINING SET

1,848

77 synthetic days

HOLDOUT: 504 OBSERVATIONS

CHRONOLOGICAL HOLDOUT MAE



All metrics are synthetic observations per hour; cannot generalize to real facilities.

FEATURES: HOUR-OF-DAY FOURIER TERMS + WEEKEND FLAG

ENGINEERING ARCHITECTURE

A local-first architecture with no deployment dependencies.



CREDIBILITY BEFORE HYPE

What we tested — and what remains unproven.

Our architecture is reproducible; operational and clinical validity require future real-world study.

VERIFIED INSIDE THE PROTOTYPE

10 automated tests: reproducibility, conservation of patient state, resource caps, matched arrivals and acuity, surge sensitivity, optimizer completeness, edge cases.

NOT CLAIMED

No real patient data; no clinical validation; no safe staffing prescription; no live hospital integration; no demonstrated real-world improvement.

NEXT: event-log calibration → prospective validation → clinical governance → operational integration

PROOF OF CONCEPT

Rehearse the Surge. Improve the Response.

2D / 3D hospital twin • seeded simulation • trained forecast • scenario optimization • tests • documentation

QUICK START

Launch Application

TEAM SynapseX • MEMBERS [Etcherla Sai Manoj , Parameswara Meena Peddakotla , Madhavaneni Vivek Rao , Valivarthi Bhargava Varma]

SOURCE: [https://github.com/saimanoj22/hospitalflow-ai] DEMO VIDEO: [https://youtu.be/gJRL3qPQtG0]